Assignment-1

Definite Integration

1. $\int_{\ln\lambda}^{\ln\left(\frac{1}{\lambda}\right)} \frac{f\left(\frac{x^2}{3}\right)(f(x)+f(-x))}{g(3x^2)\,(g(x)-g(-x))}dx =$ where $\lambda > 1$

(1) 0 (2) 1 (3) λ (4) $\frac{1}{\lambda}$

2. If $f(\pi) = 2$ and $\int_0^{\pi}(f(x)+f''(x))\sin x\,dx = 5$ then $f(0)$ is equal to

(1) 7 (2) 3 (3) 5 (4) 1

3. $\int_0^2 [x+[x+[x]]]dx =$ _____, [.] denotes GIF

(1) 0 (2) 1 (3) 2 (4) 3

4. $\int_{-2\pi}^{5\pi} \cot^{-1}(\tan x)dx =$ _____.

(1) $\frac{7\pi}{2}$ (2) $\frac{2\pi}{7}$ (3) $\frac{2\pi^2}{7}$ (4) $\frac{2\pi^2}{7}$ $\frac{7\pi^2}{2}$

5. The value of $\int_0^1 \left(\prod_{r=1}^{n}(x+r)\right)\left(\sum_{k=1}^{n}\frac{1}{x+k}\right)dx =$

(1) $n!$ (2) $(n+1)!$ (3) $n\,.\,n!$ (4) $n\,(n+1)!$

6. If $f: D \to R$ be such that $f(x) = \sqrt{\sin(\cos x)} + \log(-2\cos^2 x + 3\cos x - 1)$ then $\int_{x_1}^{x_2}\left[\cos x - \frac{1}{2}\right]dx =$ _____ ([.] is GIF) also $(x_1\ x_2) \subseteq D$

(1) 0 (2) -1 (3) 1 (4) $-\frac{1}{2}$

7. $\int_{\frac{1003\pi}{2008}}^{\frac{1003\pi}{2008}} \frac{\sqrt[7]{\sin 8x} + \sqrt[7]{\cos 8x}}{\sqrt[9]{\sec 5x} + \sqrt[9]{\operatorname{cosec} 5x}} \log\left(\frac{3+4\tan x}{3+4\cos x}\right)dx =$

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{1003\pi}{2008}$ (4) $\frac{\pi}{2008}$

8. If $f(x)$ be a continuous function for all real values of 'x' and satisfies $\int_0^x f(t)dt = \frac{x^2}{2} + \int_x^8 t^2 f(t)dt$, then $\int_{-1/2}^{1} f(x)dx =$.

(1) $\log\left(\frac{5}{8}\right)$ (2) $\frac{1}{2}\log\left(\frac{8}{5}\right)$ (3) $2\log\left(\frac{8}{5}\right)$ (4) $\frac{1}{2}\log\left(\frac{5}{8}\right)$

9. If $A=\int_{-2}^{0}\frac{|\sin x|}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}dx$ and $B=\int_{0}^{2}\frac{|\sin x|}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}dx$ where [.] is GIF then

(1) $A=B$ (2) $A=-B$ (3) $A=2B$ (4) $A=-2B$

10. $\int_{0}^{64}\{x^{1/3}\}dx=$ _____ {.} denotes fractional part of x

(1) 28 (2) 32 (3) 35 (4) 36

11. If $f(x)=\int_{0}^{1}|x-t|\,dt$, $x\in[0,1]$, then $\int_{0}^{1}f(x)dx=$

(1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

12. If $f(x)$ is monotonic and differentiable function, then $\int_{f(a)}^{f(b)}2x(b-f^{-1}(x))dx=$

(1) $\int_{a}^{b}f^2(x)dx$ (2) $\int_{a}^{b}(f^2(x)-f^2(a))dx$

(3) $\int_{a}^{b}(f^2(x)-f^2(b))dx$ (4) $\int_{a}^{b}(f^2(x)+f^2(b))dx$

13. Let $f(x)$ be differentiable function such that $f(x)=x^2+\int_{0}^{x}e^{-t}f(x-t)dt$ then $f(1)=$

(1) 1 (2) 2 (3) $\frac{1}{3}$ (4) $\frac{4}{3}$

14. $\int_{0}^{\pi/4}\frac{\sin x+\cos x}{3+\sin 2x}dx=$

(1) $ln\,3$ (2) $\frac{1}{2}ln3$ (3) $\frac{1}{3}ln3$ (4) $\frac{1}{4}ln3$

15. $\int_{1}^{e}\frac{(1+\tan x)^2dx}{1+\ln x^{x+1}+(\ln\sqrt{x})^2}=$

(1) e (2) $ln\,(e-1)$ (3) $ln\,(e+1)$ (4) $\frac{\pi}{2}$

16. $\int_{0}^{\pi/4}e^{\sec x}\frac{\sin\left(x+\frac{\pi}{4}\right)}{\cos x(1-\sin x)}dx=$

(1) $\frac{1}{\sqrt{2}}(e^{\sqrt{2}}-e)$ (2) $\left(1+\frac{1}{\sqrt{2}}\right)e^{\sqrt{2}}-\frac{e}{\sqrt{2}}$

(3) $\left(1-\frac{1}{\sqrt{2}}\right)e^{\sqrt{2}}+\frac{e}{\sqrt{2}}$ (4) $\frac{1}{2}\left(e^{\sqrt{2}}+1\right)$

17. Let $f(x)=\int_0^x \frac{dr}{\sqrt{1+r^3}}$ and $g(x)$ be the inverse function of $f(x)$ then

(1) $2g''=g^2$ (2) $2g''=3g^2$ (3) (4)

18. $\lim_{x\to\infty}(1+x)\int_0^x e^{t^2}-x^2dt=$

(1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) 2

19. $\lim_{n\to\infty}\frac{(\sqrt{1}+\sqrt{2}+\sqrt{3}+....+\sqrt{x})\left(\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+...+\frac{1}{\sqrt{n}}\right)}{1+2+3+...+n}3$

(1) (2) (3) (D)

20. The minimum value of $f(x)=\int_0^4 e^{|x-t|}dt\ 0\le x\le 4$ is

(1) 1 (2) e (3) e^2-1 (4) $2(e^2-1)$

21. $\int_0^2[\tan^{-1}x]dx+\int_0^2[\cot^{-1}x]dx$ = (where [.] is GIF)

(1) $1-\cot 2$ (2) $1+\cot 2$ (3) $2(1+\cot 2)$ (4) $2(1-\cot 2)$

22. If $f(x)=ax^2+bx+c$, $\int_0^4 f(x)dx=1$, $\int_0^4 x\ f(x)dx=2$, $\int_0^4 x^2f(x)dx=3$, then in the interval [0, 4] the equation $f(x)=0$ has

(1) No root (2) exactly one root (3) as most one root (4) atleast one root

23. Let $f(x)$ be a differentiable function. If $f(0)=0, f(1)=1$, then the minimum value of $\int_0^1 (f'(x))^2dx$ is

(1) 0 (2) $\frac{1}{2}$ (3) 1 (4) 2

24. If $f_n(x)=\frac{fofofo....of(x)}{n \text{ times}}$ and $f(x)=30-2x-x^3$. If α, β be two positive integers $\alpha<\beta$, which satisfy $f_{2018}(x)<f_{2017}(-x)$ then $\int_\alpha^\beta (f(x)+x^3)dx=$

(A) 0 (B) 8 (C) 16 (D)

25. Let $\lambda=\int_0^1\frac{1}{1+x^3}dx$, $p=\lim_{n\to\infty}\left[\frac{\sum_{r=1}^n(n^3+r^3)}{n^3n}\right]^{1/n}$ then $ln\ p=$

(A) $ln\ 2-1+\lambda$ (B) $ln\ 2-3+3\lambda$ (C) $2ln2-\lambda$ (D) $ln\ 4-3+3\lambda$

26. Let f be a polynomial function such that for all $x\in R, f(x^2+1)=x^4+5x^2+3$ and if $g(x)$ is primitive of $f(x)$ w.r.t. x and $g(0)=$ then

(A) the minimum value of $g'(x)$ is $-\frac{13}{4}$ (B) $\int_{-2}^2(2g(x)-3x^2)dx=0$

(C) $g(x) = 0$ has atleast one real root in $\left(\frac{1}{2}, 1\right)$ (D) $g(x) = x$ has exactly two real roots

27. Let $S = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx$ then which of the following is/are correct?

(A) $S < 2\left(e - \frac{1}{e}\right)$ (B) $S < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$ (C) $S = \int_0^2 t\sqrt{et}\,dt$ (D) $3 < S < 5$

28. Let f be a differentiable function on R and satisfying the equation $\int_0^x f(t)dt + \int_0^x + f(x-t)dt = -1 + e^{-x} \forall\ x \in R$ then which of the following is/are true?

(A) $f(2) = e^{-2}$ (2) $f(0) + f'(0) = 1$ (C) $f'(0) = 2$ (D) $f'(0) = 1$

29. If $\lim_{x\to\infty}\left[\frac{^{3n}C_n}{^{2n}C_n}\right]^{\frac{1}{n}} = \frac{a}{b}$ where a, b are relatively prime positive integers then $\frac{a+b+1}{a-b} =$

30. Let $f(x)$ be a continuous function defined from $[0, 2] \to R$ and satisfying the equation $\int_0^2 f(x)\,(x - f(x))dx = \frac{2}{3}$ then the value of $2f(1) =$ _____.

31. Let $A_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1} nx}{\sin^{-1} nx} dx$ and $\lim_{x\to\infty} n^2 A_n = L$ then $8L =$ _____.

32. Let $f(x)$ be A derivable function satisfying $f(x) = \int_0^x e^t \sin(x-t)$ and $g(x) = f''(x) - f(x)$. Then which of the following is not in the range of $g(x)$

(A) 0 (B) 1 (C) $\sqrt{2}$ (D) $\sqrt{3}$

33. $\int_0^\infty \frac{\tan^{-1} ax - \tan^{-1} bx}{x} dx (a > 0, b > 0)$

(A) 0 (B) $\frac{\pi}{2} ln\ ab$ (C) $\frac{\pi}{2} ln\left(\frac{a}{b}\right)$ (D) $\frac{\pi}{4} ln\left(\frac{a}{b}\right)$

34. Let $g(x)$ be inverse of $f(x)$ and $f(x)$ is given by $f(x) = \int_3^x \frac{1}{\sqrt{t^4 + 3t^2 + 13}} dt$ then $g'(0) =$

(A) $\frac{1}{\sqrt{13}}$ (B) 11 (C) 121 (D) $\frac{1}{13}$

35. If $f(x)$, $g(x)$, $h(x)$ be twice differentiable function on $[0, 2]$ satisfying $f''(x) = g''(x) = h''(x)$, $f'(1) = 2g' = h'(1) = 4$ and $f(2) = 3g(2) = h(2) = 9$ then which of the following is/are true?

(A) $\int_0^2 (f(x) + h(x) - 2g(x))dx = 16$ (B) $\int_0^2 (f(x) + h(x) - 2g(x))dx = 8$

(C) $f(x) - g(x) = 2x^2 + 2$ has two real roots (D) $f(x)$ and $h(x)$ are identical function

36. If $L = \lim_{x\to\infty} \frac{x^3(e^{1/n} + e^{2/n} + ... + e)}{(n+1)^m\ (1^m + 4^m + ... + n^{2m})}$ is non-zero finite real number then, which of the following is/are true?

(A) $L = 3(e-1)$ (B) $L = e-1$ (C) $m = 1$ (D) $m = \frac{4}{3}$

37. If $I = \int_0^{\pi/2} e^{-\lambda \sin x}\, dx,\ \lambda \in (0,\infty)$ then which of the following is/are true?

(A) $I < \frac{\pi}{2}$ (B) $I > \frac{\pi}{2}(e^{-\lambda}+1)$ (C) $I > \frac{\pi}{2}e^{-\lambda}$ (D) $I > 0$

38. Which of the following holds good for a real valued function defined by

$$f(x) = x^2 + x^2\int_{-1}^{1} t\ f(t)dt + x^3\int_{-1}^{1} f(t)dt$$

(A) $\int_{-1}^{1} t\, f(t)dt = \frac{10}{11}$ (B) $f(1)+f(-1) = \frac{30}{11}$ (C) $\int_{-1}^{1} t\ f(t) < \int_{-1}^{1} f(t)dt$ (D) $f(1)-f(-1) = \frac{20}{11}$

39. Let $f: [0, \infty) \to R$ be continuously strictly increasing function such that $f^3(x) = \int_0^x t\, f^2(t)dt$ for every $x > 0$, then which of the following is/are correct?

(A) $f(6) = 6$ (B) $f(x$ is onto

(C) $\int_0^1 I(x)dx = \frac{1}{18}$ (D) Number of solution of $f(x) = 6$ are two

Passage (40-41)

Let $f_n(x) = \sum_{r=1}^{n} \frac{\sin^2 x}{\cos^2\left(\frac{x}{2}\right) - \cos^2\left(\frac{2r+1}{2}\right)x}$ $g_n(x) = \sum_{r=1}^{n} f_r(x)\ \forall\ n = 1,2,3.....$

40. If $I_n = \int_0^{\pi} \frac{f_n(x)}{g_n(x)}dx$ then $\sum_{k=1}^{100} I_k =$

(A) 25 π (B) 50 π (C) 75 π (D) 100 π

41. The value of $\lim_{x\to 0} \int_0^x \frac{dt}{x f_9(t)\ g_9(t)} =$

(A) $\frac{50}{9}$ (B) $\frac{10}{9}$ (C) $\frac{100}{9}$ (D) $\frac{25}{9}$

42. $I_{m,n} = \int_0^1 x^m (\ln x)^n dx =$

(A) $-\frac{m+1}{n+1} I_{m,\,n+1}$ (B) $-\frac{m}{n+1} I_{m,\,n-1}$ (C) $-\frac{n}{m+1} I_{m,\,n-1}$ (D) $\frac{m}{n+1} I_{m,\,n-1}$

43. Which of the following inequalities is/are not incorrect?

(A) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{3}}{8}$ (B) $2e^{-1/4} < \int_0^2 e^{x^2-x}\, dx < 2e^2$

(C) $\frac{1}{3} < \int_0^1 x(\sin x + \cos x)^2 dx < \frac{1}{2}$ (D) $\frac{1}{2} < \int_0^2 \frac{dx}{2+x^2} \le \frac{5}{6}$

44. $\lim_{n\to\infty} n^2 \int_{-1/n}^{1/n} (2015 \sin x + 2016 \cos x)\,|\,x\,|\,dx = \lambda$, then the sum of digits of λ is _____

45. $\int_0^{\pi} \sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$ has the value equal to $\frac{\pi}{k}+\sqrt{w}$, where k and w are positive integers. Then sum of digits of HCF of kw and w^k is

46. The value of a positive integer $n \le 5$ such that $\int_0^1 e^x(x-1)^n\,dx = 16-6e$ is

47. Let a, b, c be positive constants. When the value of integral $\int_0^1 (ac\,x^{b+1}+a^3\,bx^{3b+5})dx$ is independent of b, then $a^p c = \frac{r}{q}$, where p, q, r are expressed in simplest form, then $p+q+r =$

48. Match the following: $a < b$; $a, b \in N$

Column I	Column II
(A) $\int_a^b \sqrt{\frac{x-a}{b-x}}dx$	(P) $\frac{\pi}{\sqrt{ab}}$
(B) $\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}}dx$	(Q) $(b-a)\frac{\pi}{2}$
(C) $\int_a^b \frac{1}{x\sqrt{(x-a)(b-x)}}dx$	(R) $(a+b)\frac{\pi}{2}$
(D) $\int_a^b \frac{x}{\sqrt{(x-a)(b-x)}}dx$	(S) π

49. The value of $\int_0^1 \frac{(1-x^2)^{1/3}\,dx}{(1-x^3)^{1/2}\,dx}$ is equal to

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{6}$

50. Let the function $f(x)$, $g(x)$ be differentiable and strictly increasing $\forall\, x \in R$. Also $f(0) = 2$, $f(1) = 3$ and $f'(x)$ is continuous. If $f^2(x) = 1 + g^2(x)$ and $J = \int_0^1 \left(\frac{g(x)}{f(x)}\right)^1 \frac{1}{g(x)}dx$ then the value of $6J + 1$ is

(A) $ln\left(\frac{3}{2}\right)$ (B) $ln\left(\frac{27}{8}\right)$ (C) $ln\left(\frac{9}{4}\right)$ (D) NOT

51. If $I_1 = \int_0^{\pi/4} \frac{1}{(1+\tan x)^2}dx$ and $I_2 = \int_6^{\pi/4} (1+\tan x)^2\,dx$ then $\frac{I_2}{I_1} =$

(A) 1 (B) 2 (C) 3 (D) 4

52. The value of the definite integral $\int_0^{\pi} ((1-x\sin 2x)e^{\cos^2 x} + (1+x\sin 2x)e\sin^2 x)dx$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

53. Which of the following statements is/are false?

(A) $\int_1^4 e^{x^2}\,d(x^2) = e^{16} - e^1$

(B) If $0 < x < \frac{\pi}{2} < \int_{\frac{1}{2}}^{\sqrt{3}/2} ln(e^{\cos x}) d(\sin x) = \frac{\pi}{12}$

(C) Let $f(x) = \int_0^x |\cos t|\, dt$, then $f'(x)$ is continuous but not differentiable at $x = \pi/2$

(D) Let $f(x) = \int_{-2}^x |t+1|\, dt$, $x \in \left[-\frac{3}{2}, 1\right]$. Then f is continuous but not differentiable at $x = -1$

54. Which of the following statements is/are true?

(A) Let f be strictly increasing continuous function defined on the set of real numbers. If $f(1) = 2, f(2) = 4$ then $\int_1^2 f(x)dx$ can be equal to $\frac{25}{8}$.

(B) Let $f : R^+ \to R^+$ is such that $f''(x) \geq 0 \ \forall \ x \in [a, b]$. Then the value of $\int_a^b f(x)dx$ can not exceed $\frac{(f(a)+f(b))(b-a)}{2}$

(C) The value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} < \frac{\pi}{4}$

(D) The value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} > \ln 2$

55. Let $f : R \to (0, \infty)$ be a real valued function such that $\int_0^x t\, f(x-t)dt = e^{2x-1}$, then which of the following is/are correct?

(A) $f(0) = 4$ (B) $(f^{-1})^1(4) = \frac{1}{8}$ (C) $\lim_{x\to 0} \frac{f(x)-4}{x} = 4$ (D) $\left(\frac{df}{d(e^x)}\right)_{at\ x=0} = 8$

Passage: (56-57)

Let f be a function defined by $f(k) = \int_0^{\pi/2} |k - 2\sin t)\cos t\ dt$ where k is real parameter. Then

56. Which of the following statements is correct?

(A) $f(k)$ is continuous $\forall\ k$ but not derivable at two value of k.

(B) $f(x)$ is differentiable $\forall\ k$

(C) $f(t) = 1 - k \ \forall\ k \geq 2$

(D) $f(k) = k - 1 \ \forall\ k \geq 0$

57. The minimum value of $f(t)$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 0 (D) 1

Passage(58-60)

Let $f(x)$ and $g(x)$ be two differentiable function on R satisfying $f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)dt$ and $g(x) = x - \int_0^1 f(t)dt$.

then

58. The value of definite integral $\int_0^1 f(t)dt$ lies in the integral

(A) $\left(0,\frac{1}{2}\right)$ (B) $\left(\frac{1}{2},1\right)$ (C) $\left(1,\frac{4}{3}\right)$ (D) $\left(\frac{4}{3},\frac{5}{3}\right)$

59. Minimum vertical distance between two curves $y = f(x)$ and $y = g(x)$ is

(A) $\frac{7}{3}$ (B) $\frac{1}{6}$ (C) $\frac{8}{3}$ (D) $\frac{7}{6}$

60. If the distance of the point $p(x_1, y_1)$ lying on the curve $y = f(x)$, from the curve $y = f(x)$ is least, then $x_1 =$

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{7}{6}$

61. Let $y = p(x)$ be a quadratic function with $p'(2) = 1$, then the vale of $\int_{2-\pi}^{2+\pi} p(x)\sin\left(\frac{x-2}{2}\right)dx$ is _____

62. If $\int_0^{2\pi} \frac{dx}{2+\sin 2x} = k\int_0^{\pi/2} \frac{dt}{7+\cos 2t}$, then the value of $\frac{k}{4}$ is _____

63. Let f be twice differentiable function satisfying $\frac{e^x f''(x) - e^x f'(x)}{e^{2x}} = t$ and $f(0) = 0, f'(0) = 1$, then the value of $\left[\int_0^1 \frac{f(x)}{x}dx\right]$ (where [.] is GIF) is

64. Let $Q \in \left(0,\frac{\pi}{2}\right)$ then the value of $\lim_{a\to 0} \frac{1}{a^3}\int_0^a \ln(1+\tan a \tan x)dx$ is equal to $\frac{1}{k}$ where k is _____.

65. Suppose $f(x)$ is a continuous function such that $f(x) = \int_0^x f(t)dt$, then $e^{-4}f(4) =$ _____.

66. Let $f(x)$ be a periodic function with F.P. 12. If $\int_{-6}^{12} f(x)dx = 9$, $\int_0^{12} f(x)dx = 12$ then the value of $\frac{1}{3}\left|\int_0^6 f(x)dx\right|$ is _____.

67. Using $\int_0^1 x^n\,dx = \frac{1}{n+1}, n \neq -1$ the sum of the series $1-\frac{1}{5}+\frac{1}{7}-\frac{1}{11}+\frac{1}{13}-\frac{1}{17}+\frac{1}{19}....+....\infty = \frac{\pi}{k\sqrt{3}}$ where k is equal to

68. The value of $\int_0^\infty \frac{\ln x}{x^2+2x+4}dx$ equals to

(A) $\frac{\pi}{3\sqrt{3}}\ln 2$ (B) $\frac{\pi}{\sqrt{3}}\ln 2$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{2\sqrt{2}}\ln 2$

69. Consider a real valued continuous function f such that $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + tf(t))dt$ then the difference of maximum, minimum value of f in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ is

(A) $\pi + 1$ (B) $2(\pi + 1)$ (C) π (D) can't be determined

70. The value of $\int_{-1}^1 \frac{2x^{1007} + x^{3023} + 4x^{2016}(\tan^{-1}x)^{2015}}{1+x^{2016}}dx$

(A) $\frac{\pi}{2} ln\, 2$ (B) $\pi ln\, 2$ (C) $\frac{\pi}{8} ln\, 2$ (D) 0

71. The value of $\int_{-1}^{1} \frac{2x^{1007} + x^{3023} + 4x^{2016}(\tan^{-1} x)^{2015}}{1+x^{2016}}$

(A) 0 (B) $\frac{1}{504} - \frac{\pi}{4}$ (C) $\frac{1}{504} + \frac{\pi}{2016}$ (D) $\frac{\pi}{504} + \frac{1}{4}$

72. $\lim_{n\to\infty} ({}^{2n}C_n)\frac{1}{n} =$

(A) 1 (B) 4 (C) e^2 (D) e^4

73. Consider the function $\begin{cases} \int_0^x (4+|t-2|)dt, & x>3 \\ dx^2 + bx, & x \le 3 \end{cases}$ and if $f(x)$ is differentiable at $x = 3$ then

(A) $27\, ab = 1$ (B) $f(1) = \frac{85}{18}$ (C) $f(4) = 20$ (D) $\int_4^6 (f'(x) + 2)dx = 18$

74. Let $f(x)$, $g(x)$ be any two continuous functions, a is non-zero real number b is positive real number but not equal to 1 then the value of $\int_{-a}^{a} \frac{f(x)}{b^{g(x)}+1} dx$ is

(A) independent of f, if f even, g is odd (B) independent of g, if f even, g is odd

(C) independent of f, g if both f, g are even (D) independent of f, g if both f, g are odd

75. Let $f = [0, \infty) \to R$ be a continuous strictly increasing function, such that $f^3(x) = \int_0^x t\, f^2(t)dt \;\forall\; x \ge 0$, then which of the following is/are true

(A) $f(x)$ is an onto function

(B) $\int_0^1 f(x)dx = \frac{1}{18}$

(C) Number of solutions of $6f(x) = e^x$ is 2

(D) The graph of $y = 6f(x)$ intersects the graph of the curve $y = 3x^2 + 2x + 2$ at one point.

76. If $\int_{\sin x}^{1} t^2 f(t)dt = 1 - \sin x$ where $x \in \left(0, \frac{\pi}{2}\right)$ then $f\left(\frac{1}{\sqrt{2}}\right)$ is equal to

77. If $\int_2^8 \frac{dx}{x\sqrt{(x-2)(8-x)}} dx = \frac{\pi}{k}$ then the absolute value of k is

78. The value of $\int_{\infty}^{a} \frac{\sin^{-1} e^x + \sec^{-1} e^{-x}}{(\tan^{-1} e^a + \tan^{-1} e^x)(e^x + e^{-x})} dx \;\; a \in R$, is

(A) independent of a (B) depends on a (C) $\frac{\pi}{2} ln\, 2$ (D) $\frac{\pi}{2} ln(2 + \tan^{-1} e^a)$

79. If $I_n = \int_0^1 (1 + x + x^2 + ... + x^{n-1})(1 + 3x + 5x^2 + ... + (2n-3)x^{n-2} + (2n-1)x^{n-1})\, dx \;\forall\; n \in N$ then

(A) $I_3 + I_4 = I_5$ (B) I_n is a perfect square

(C) $I_{n+1} - I_n = 2n + 1 \;\forall\; n \in N$ (D) $I_{n+1} - I_n = 2x - 1 \;\forall\; n \in N$

80. The value of $L=\lim_{n\to\infty}\dfrac{\left(\sum_{r=1}^{2n}\sqrt{r}\right)\left(\sum_{r=1}^{3n}\dfrac{1}{\sqrt{r}}\right)}{\left(\sum_{r=1}^{5n}r\right)}$ is less than

(A) $\sqrt{3}$ (B) 2 (C) $\dfrac{32}{75}$ (D) $\sqrt{2}$

81. Let $f(x)=\int_0^x e^t\sin(x-t)dt$ then

(A) $f''\left(\dfrac{\pi}{2}\right)-f'\left(\dfrac{\pi}{2}\right)=-1$ (B) $f''(0)-f'(0)=1$

(C) $(f''(x)-f(x))\max=\sqrt{2}$ (D) $f''\left(\dfrac{\pi}{2}\right)-f'\left(\dfrac{\pi}{2}\right)=1$

Passage (81-82)

Let $f: R\to R$ be a differentiable function such that $f(x)=x^2+\int_0^x e^{-t}f(x-t))dt$. then

81. $\int_0^1 f(x)dx=$

(A) $\dfrac{1}{4}$ (B) $-\dfrac{1}{12}$ (C) $\dfrac{5}{12}$ (D) $\dfrac{12}{7}$

82. $f(x)$ is

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (D) neither one0one nor onto

Passage (83-84)

Let $I_1=\int_1^2\tan^{-1}x^2dx$ and $\int_1^2\left(\cot^{-1}x^2+\tan^{-1}\dfrac{1-x^2}{1+x^2}\right)dx=I_2$ and if $aI_1+bI_2=\dfrac{3\pi}{4}$ and also

$I_1=2\tan^{-1}a_1-\dfrac{\pi}{4}-\dfrac{1}{\sqrt{2}}\left(\tan^{-1}\dfrac{3}{2\sqrt{2}}+\ln\left(\dfrac{a_2+a_3\sqrt{2}}{a_4}\right)^{1/2}\right)$ (here $a, \in I$, a_2, a_3, a_4 are coprimes, a, b are real numbers) then

83. $a_1+a_2-a_3-a_4=$

(A) 0 (B) 7 (C) 5 (D) 11

84. $a+b=$

(A) 7 (B) 3 (C) -2 (D) 1

85. **Column I** **Column II**

(A) If $f(x)=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2x^2}, x>0$ then (1) $\frac{\pi}{2}$

value of $f(1)$ is

(B) The value of $\int_0^{1/\sqrt{2}}\frac{x^2}{\sqrt{1-x^2}\,(1+\sqrt{1-x^2})}dx$ is (2) $-\frac{\pi}{2}$

(C) Let $\int_0^{f(x)} t^2dt = x\cos\pi x$ then the value of (3) $\frac{\pi}{4}$

$\frac{\pi^2}{8}\int_{f^2(9)f'(9)}^{1}|\sin\pi x|\,dx$ is

(D) Value of $\int_{-\frac{1}{2}}^{\frac{1}{2}}(\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x))dx$ (4) $\frac{\pi}{4}-\frac{1}{\sqrt{2}}$

86. **Column I** **Column II**

(P) the value of $\int_0^1\frac{1}{\sqrt{x}+\sqrt[3]{x}}dx$ (1) 4

$= a + b\ ln\ 2$

(Q) If $\int_0^{ln10}\frac{e^x\sqrt{e^x-1}}{e^x+8}dx = a-b\pi$ (2) 0

then $\frac{a}{b}$ = (a, b are relatively prime)

(R) $\frac{\int_0^{\pi/2}(\cos x)^{\sqrt{2}+1}dx}{\int_0^{\pi/2}(\cos x)^{\sqrt{2}-1}dx} = a-\sqrt{b}$ then $|a-b|$ (3) 1

(S) $\int_0^2\frac{\ln(1+2x)}{1+x^2}dx = (\tan^{-1}a)\ln\sqrt{b}$ (4) 3

then $|b-a|$